Score for this attempt: **4** out of 4

Question 1

1 / 1 pts

We roll two fair dice and denote the sum of the two numbers by X. What is the size of the event $\{X = x\}$ if x =5, and we use the sample space $\{(a_1, a_2) : 1 \leq a_1, a_2 \leq 6\}$.

(All the variables are assumed to be integers.)

$\{X=5\} = \{(1,4), (2,3), (3,2), (4,1)\}$

4

Question 2

1 / 1 pts

Choose the statements that are true for any $(\Omega, \mathcal{F}, \mathbb{P})$ probability space.

- [x] If A is an event and $P(A) = 1/3$ then $P(A^c) = 2/3$.
- [ ] If A and B are events and $A \cup B = \Omega$ then $P(A) + P(B) = 1$.

  False because we may have $P(A \cap B) > 0$, in which case $P(A) + P(B) > 1$.
- [x] If A and B are events then $P(A \cup B)$ is always at least as large than $P(A)$.
- [ ] If A is an event and $P(A) = 0$ then $A = \emptyset$.

  False. For a counterexample, suppose we choose x uniformly at random from $[0,1]$. Take A to be a singleton set, e.g. $A = \{1/2\}$.

Question 3

1 / 1 pts

On each round, two individuals separately flip a fair coin. If one individual flips heads and the other flips tails, then the game ends and the person who flips heads wins. However, if these individuals both flip heads or both flip tails, then the game continues and they must start over. What is the probability that the game ends after exactly 6 rounds?

Provide a numerical answer with precision up to 3 decimal places.

0.016

Let X be the # of before the game ends.

In each round, there are 4 possible ordered pairs of rolls for player 1 and player 2, with each ordered pair equally likely. These pairs are: (H,H), (H,T), (T,H), (T,T). Half these pairs lead to another round, i.e. if the game had k-1 rounds, the probability it continues another round is $\frac{1}{2}$. This means that

$$P(\{X=k\}) = \left(\frac{1}{2}\right)^{k-1} \times \left(\frac{1}{2}\right)$$

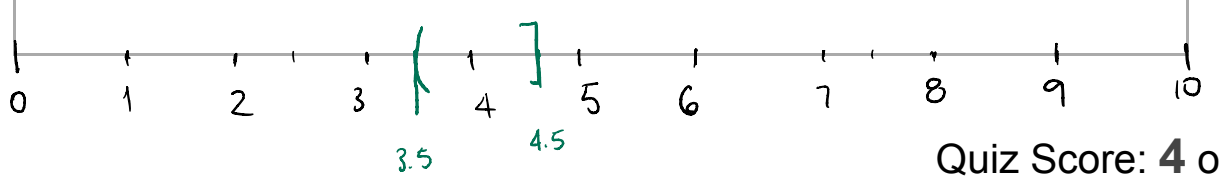
$$P(\{X=6\}) = \left(\frac{1}{2}\right)^{6}$$

Question 4

1 / 1 pts

Consider an experiment where we uniformly chose a point in the interval [0,10]. Let X be the closest integer to the chosen point, with ties going to the smaller integer. What is the probability that X = 4?

0.1



Quiz Score: **4** out of 4

$$\{X=4\} = (3.5, 4.5]$$

$$P(\{X=4\}) = \frac{\text{length}((3.5, 4.5])}{\text{length}([0,10])} = \frac{1}{10} = \boxed{0.1}$$